GOOGLE DATA ANALYTICS

CYCLISTIC BIKE SHARE

CAPSTONE PROJECT

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28/4/2026

OUTLINE

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- Background
- Business Task & Analysis Questions
- Dataset
- Tools
- Process
- Analysis
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EXECUTIVE SUMMARY

CYCLISTIC

OBJECTIVE

CYCLISTIC WANTS TO BUILD A MARKETING STRATEGY THAT TURNS CASUAL RIDERS INTO ANNUAL MEMBERS. TO DO THIS, WE NEED TO UNDERSTAND HOW MEMBERS AND CASUAL RIDERS USE THE BIKES DIFFERENTLY — ESPECIALLY WHY THEY RIDE. THIS IS BASED ON OVER 10 MILLION RIDES RECORDED FROM 2024 TO EARLY 2026.

KEY INSIGHTS

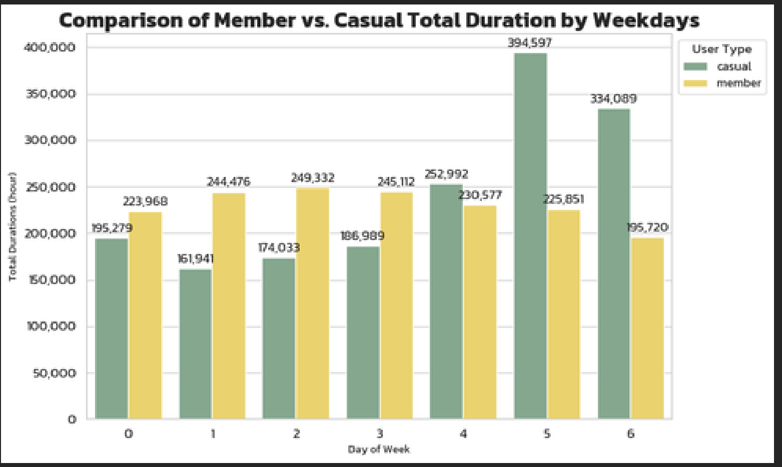
AFTER ANALYZING THE DATA, HERE ARE THE KEY DIFFERENCES BETWEEN THE TWO RIDER TYPES:

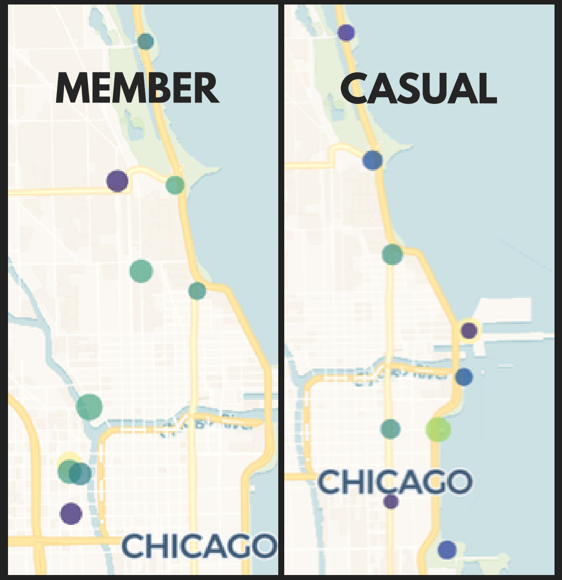
MEMBER	CASUAL
Rides more often in every season, but for shorter trips. They only ride longer when the weather is bad for outdoor activities.	Rides less often, but for much longer trips — mainly when the weather is good for being outside, from late spring to early fall.
Rides mainly to commute to work or school. Trips are routine. Start and end stations are usually in office and school areas further inside the city.	Rides mainly for tourism and leisure. Start and end stations are usually in tourist and leisure areas along the lakefront.
Most active at the start and end of the workday (rush hours). Rides more on weekdays (except Friday, which is slightly lower).	Most active from midday to evening. Rides more on weekends.

year	member_casual	ride_count	duration	
2024	casual	2086078	53948419.41	2024
	member	3642591	47325604.93	
2025	casual	1920461	45151015.34	2025
	member	3485102	43801273.95	
2026	casual	149155	2895745.01	2026
	member	490942	5775169.72	

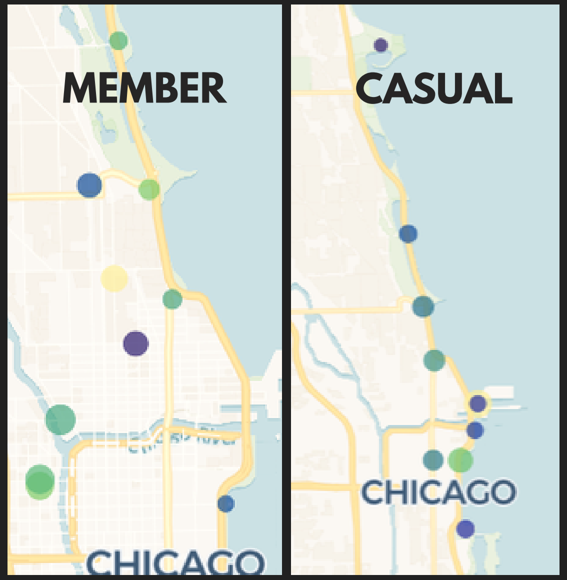
Yearly comparison of ride count and ride time. Members ride more often, but casual riders spend more time per ride.

The chart clearly shows that casual riders are very active on weekends, but drop sharply on regular workdays.





Map of start stations for both rider types



Map of end stations for both rider types

RECOMMENDATIONS

- **Flexible Membership :** Add short-term and seasonal passes (e.g. Summer Pass) to attract casual riders.
- **Contextual Marketing :** Use digital media to reach the right group by time, place, and travel season.
- **Commuter Incentives :** Give perks and reward points to commuters to build loyalty.
- **Strategic Rebalancing :** Move bikes to match demand (business areas on weekdays, tourist spots on weekends).

CYCLISTIC BACKGROUND

Cyclistic is a bike-share program with over 5,800 bikes and more than 600 docking stations. It offers many options, such as electric bikes and classic bikes. About 30% of riders use the bikes to commute to work. Its marketing strategy focuses on building awareness and flexible pricing to attract both casual riders and annual members. The main goal is to turn casual riders into annual members. The marketing analytics team wants to understand the differences between rider groups and analyze ride data to shape a new strategy.

BUSINESS TASK

" Design a marketing strategy to turn casual riders into annual members. "

ANALYSIS QUESTIONS

Three questions that guide the direction of the marketing campaign

- How do annual members and casual riders use Cyclistic bikes differently?
- Why would a casual rider decide to buy a Cyclistic annual membership?
- How can Cyclistic use digital media to encourage casual riders to become members?

DATASET

Cyclistic's historical ride data, from January 2024 to March 2026, can be downloaded from divvy_tripdata

This public dataset is based on real-world business usage. Privacy is strictly protected — the data is not linked to any rider's personal information.

Index of bucket "divvy-tripdata"

Name	Date Modified	Size	Type
202004-divvy-tripdata.zip	Jun 1st 2020, 09:50:06 pm	3.32 MB	ZIP file
202005-divvy-tripdata.zip	Jun 1st 2020, 09:50:09 pm	7.99 MB	ZIP file
202006-divvy-tripdata.zip	Jul 6th 2020, 07:31:49 am	14.73 MB	ZIP file
202007-divvy-tripdata.zip	Aug 12th 2020, 09:10:49 am	23.62 MB	ZIP file
202008-divvy-tripdata.zip	Sep 4th 2020, 10:11:40 pm	27.86 MB	ZIP file

Description : website interface

12,069,836 rows and 13 columns. Shows the bike type used, start and end stations with coordinates, and member type. ride_id is the primary key.

```
RangeIndex: 12069836 entries, 0 to 12069835
Data columns (total 13 columns):
 #   Column              Dtype
---  ------              -----
 0   ride_id             str
 1   rideable_type       str
 2   started_at          str
 3   ended_at            str
 4   start_station_name  str
 5   start_station_id    str
 6   end_station_name    str
 7   end_station_id      str
 8   start_lat           float64
 9   start_lng           float64
 10  end_lat             float64
 11  end_lng             float64
 12  member_casual       str
dtypes: float64(4), str(9)
memory usage: 2.6 GB
```

Description : Data information using info() on pandas

TOOLS

PROCESS

The data preparation process is as follows:

Format time columns correctly (String to Datetime).

Add useful columns for analysis, such as Chicago holidays and seasons.

Reorder columns for easier use, then export to a parquet file for analysis in the next file.

Remove unusable data caused by system errors, such as duplicate ride_ids and rows where the start time is after the end time.

Calculate ride duration and remove rides that are too short, to make sure the remaining data comes from real rides.

Handle missing values (Null) as follows:

- 2,206,194 rows had no station name but had coordinates. These were classified as On-street parking, since the bikes can be parked anywhere using IoT and GPS.
- 13,295 rows had missing coordinates. If a station name existed, its coordinates were filled in; if not, the row was deleted because it could not be identified.

ANALYSIS

MEMBER VS. CASUAL

Yearly comparison of ride count and ride time.

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2024	casual	2086078	53948419.41
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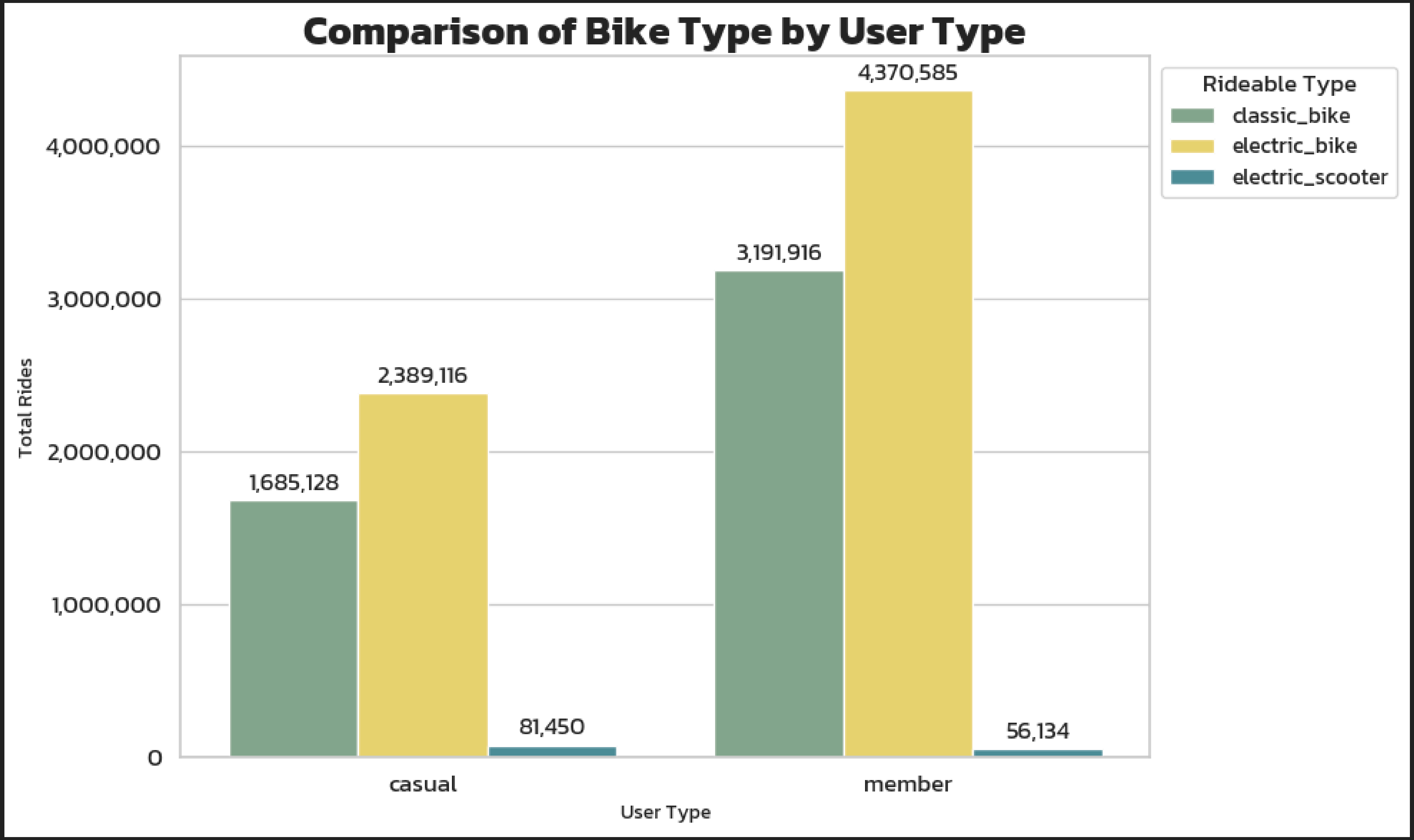
2024
2025
2026

Members ride over 1.5 times more often than casual riders, but their ride time is shorter (except in 2026, where the data only goes up to March).

RIDEABLE TYPE USED

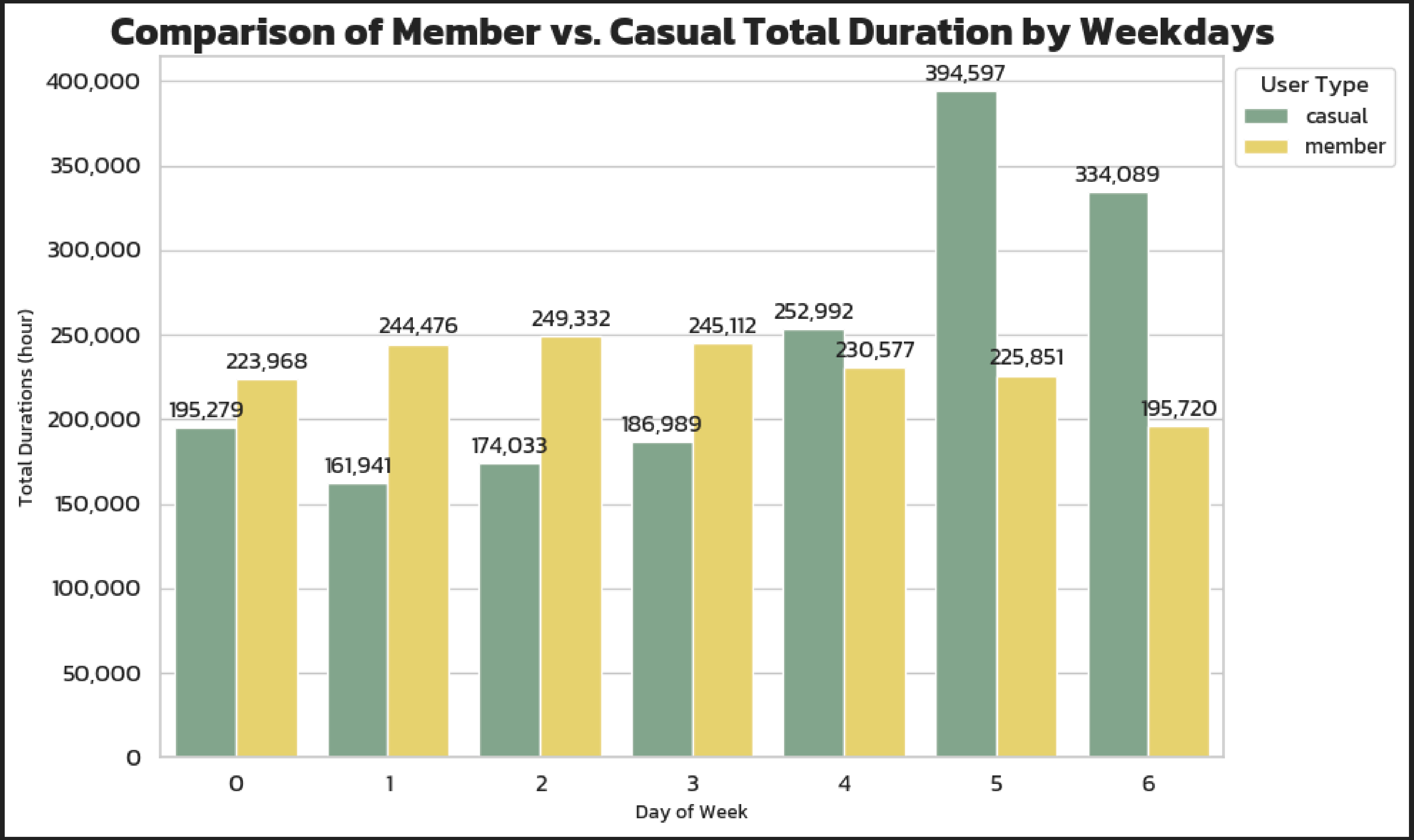


Electric bikes are the most popular with both rider types, followed by classic bikes.

Members use the main bike types much more often than casual riders.

Meanwhile, scooters are 1.6 times more popular with casual riders than with members.

WEEKDAY DURATION MEMBER VS. CASUAL



Looking at daily ride time across the week:

Members are steady and higher than casual riders on weekdays, except Friday, and drop to their lowest on weekends.

In contrast, casual riders are much higher than members on weekends.

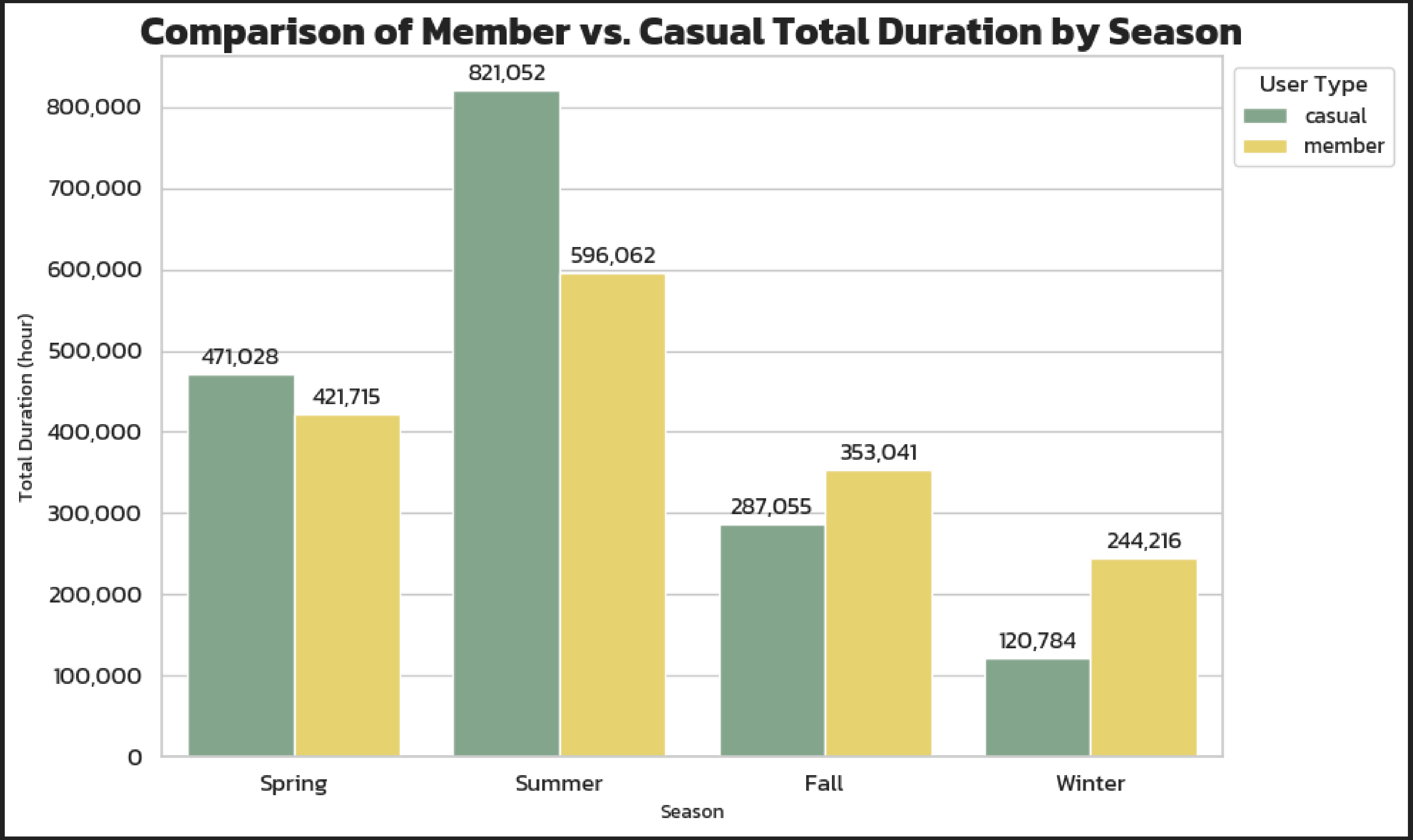
HOLIDAYS AVERAGE DURATION

holiday_name	Avg. Duration (min)
Independence Day	24.71
Labor Day	23.33
Memorial Day	22.72
Christmas Day	20.90
Martin Luther King Jr. Day	20.38
Juneteenth National Independence Day	19.45
New Year's Day	18.47
None	16.84
Columbus Day	16.21
Washington's Birthday	14.84
Thanksgiving Day	13.71
Casimir Pulaski Day	13.20
Veterans Day	12.47
Lincoln's Birthday	12.20
Election Day	10.84

> AVG

Chicago has 7 special holidays. On these days, the average ride time is higher than on non-holidays, as shown in the table.

SEASON DURATION
MEMBER VS. CASUAL



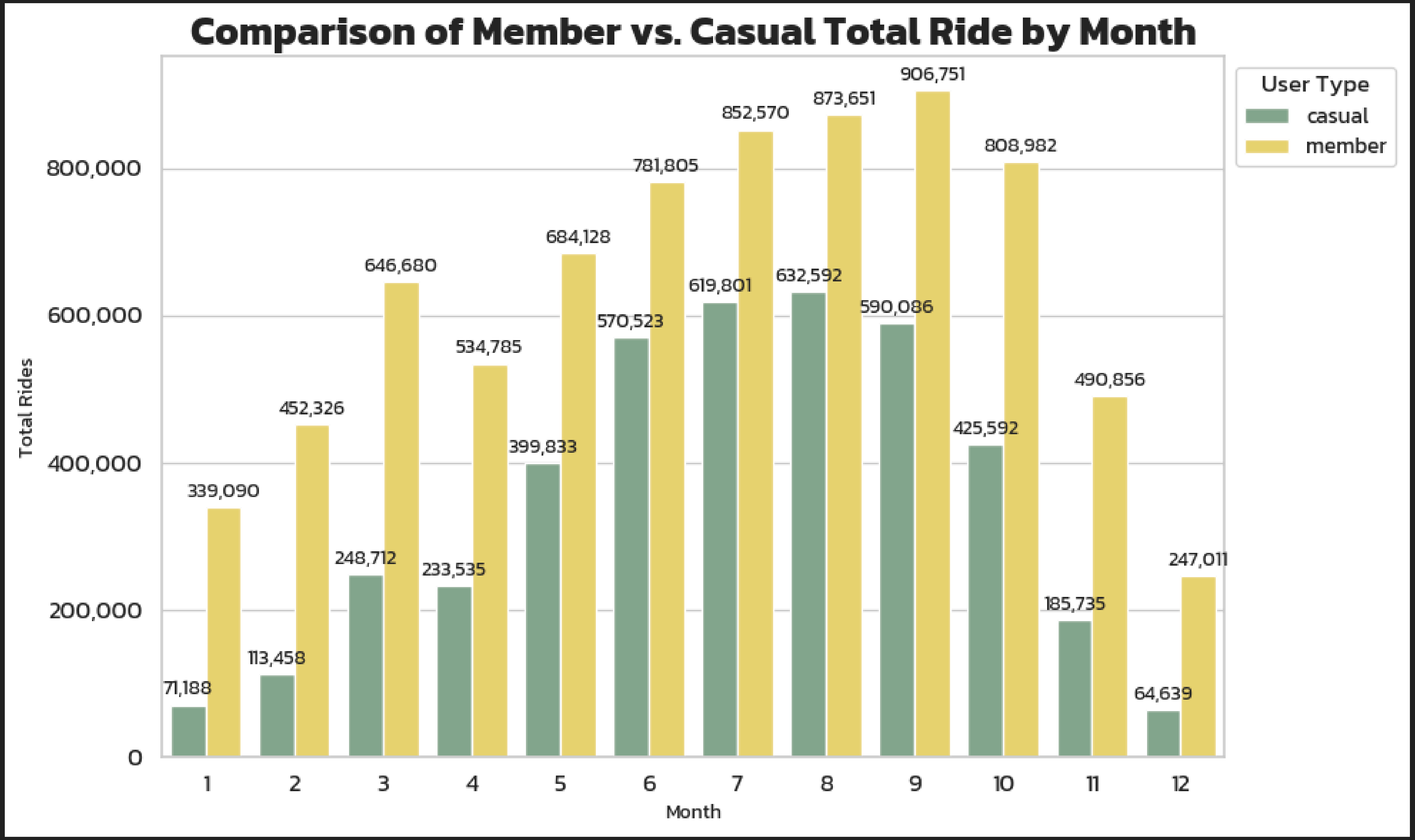
Seasons

- Spring : 20 March - 20 June
- Summer : 21 June - 21 September
- Fall : 22 September - 21 December
- Winter : 22 December - 19 March

By season, casual riders have higher ride time than members during travel-friendly seasons.

During seasons that are not good for travel, casual ride time drops sharply, while member ride time stays more stable.

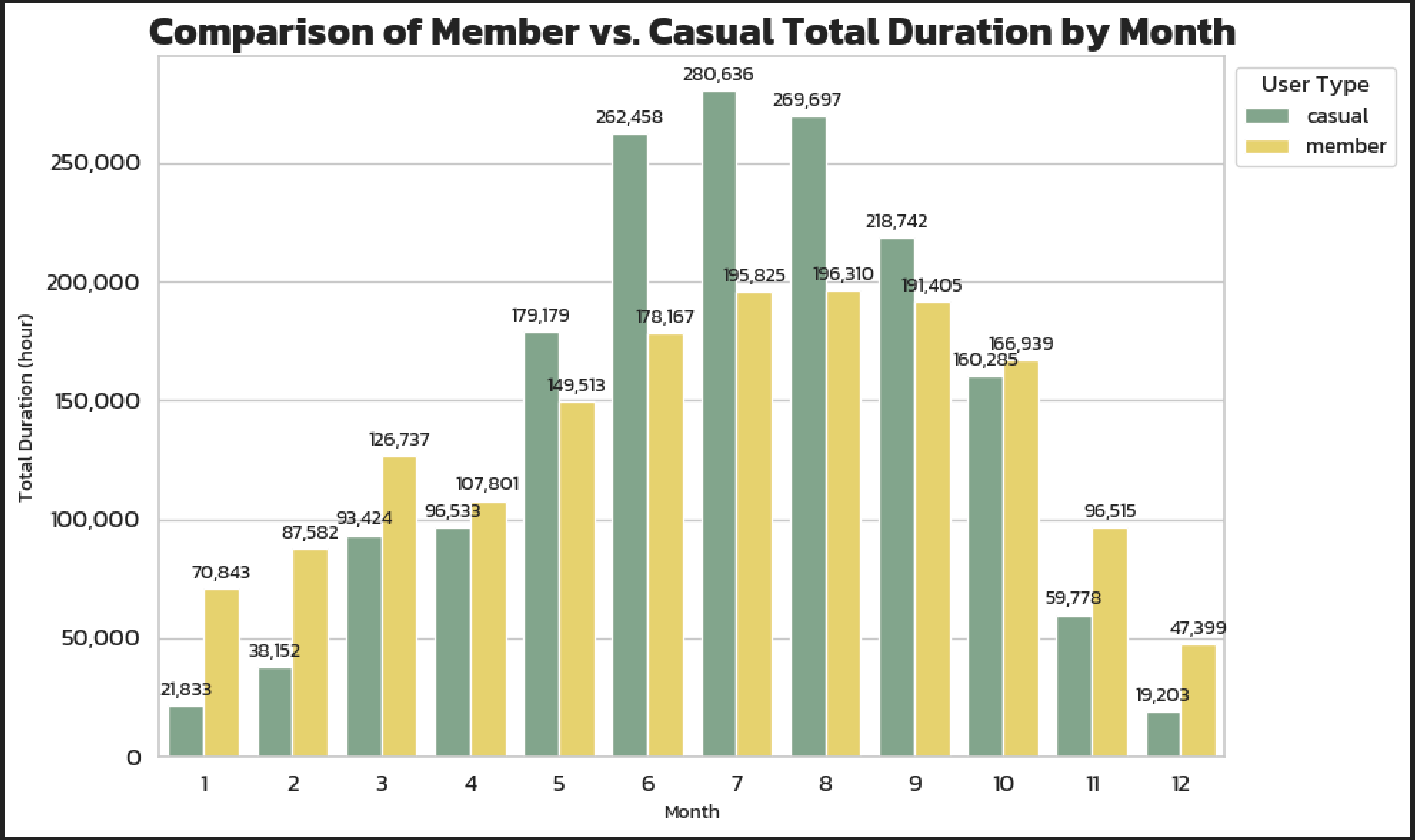
MONTHLY TOTAL RIDES MEMBER VS. CASUAL



Monthly ride count shows that members ride more often in every month.

In months with bad weather for being outside, casual rides drop sharply, while member rides stay more stable.

MONTHLY DURATION MEMBER VS. CASUAL

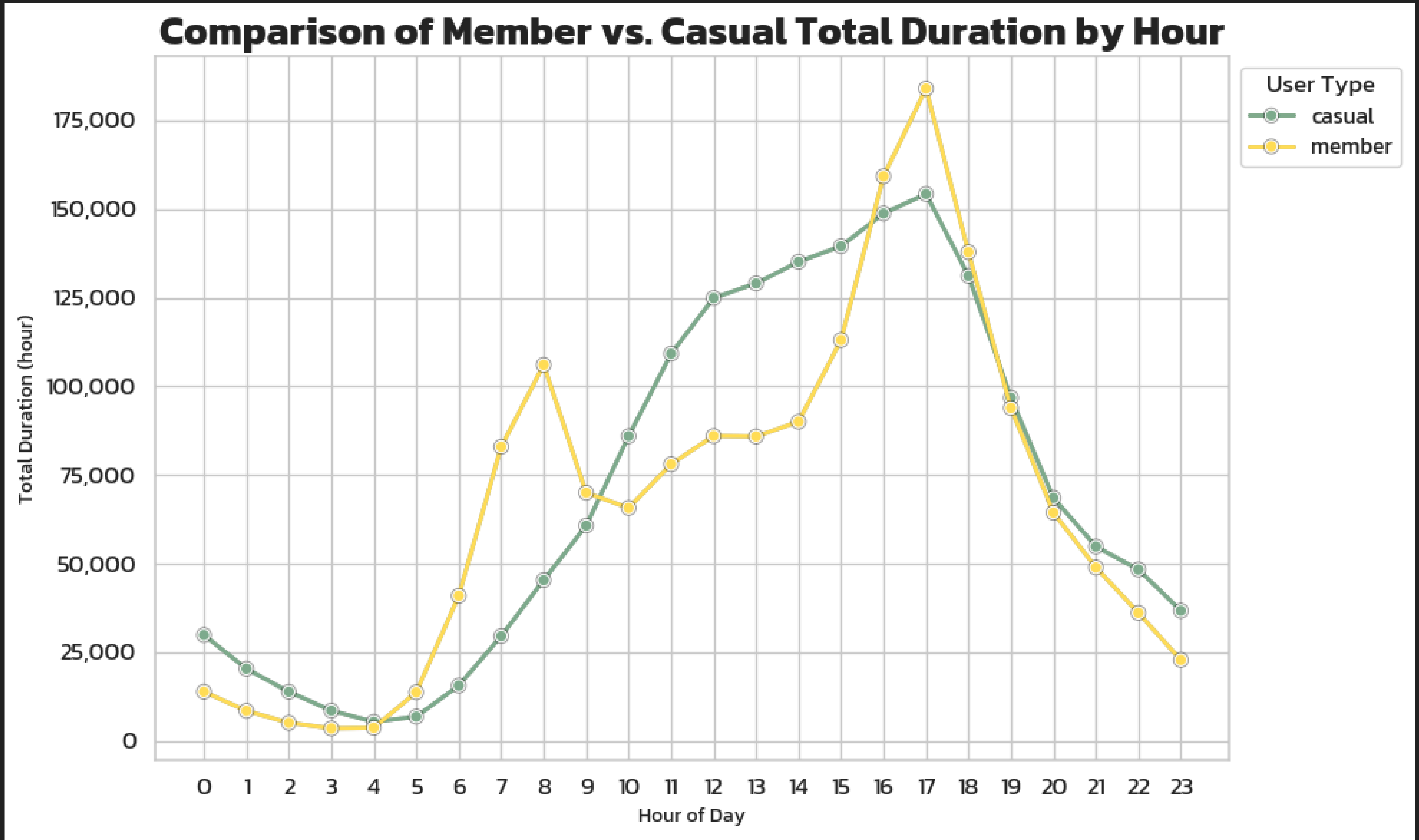


Monthly ride time shows that from May to September, casual riders ride more than members.

This is when the weather in Chicago is good and great for traveling.

This period spans late spring, summer, and early fall.

HOURLY DURATION MEMBER VS. CASUAL



Hourly ride time:

Members have sharp peaks at the start and end of the workday.

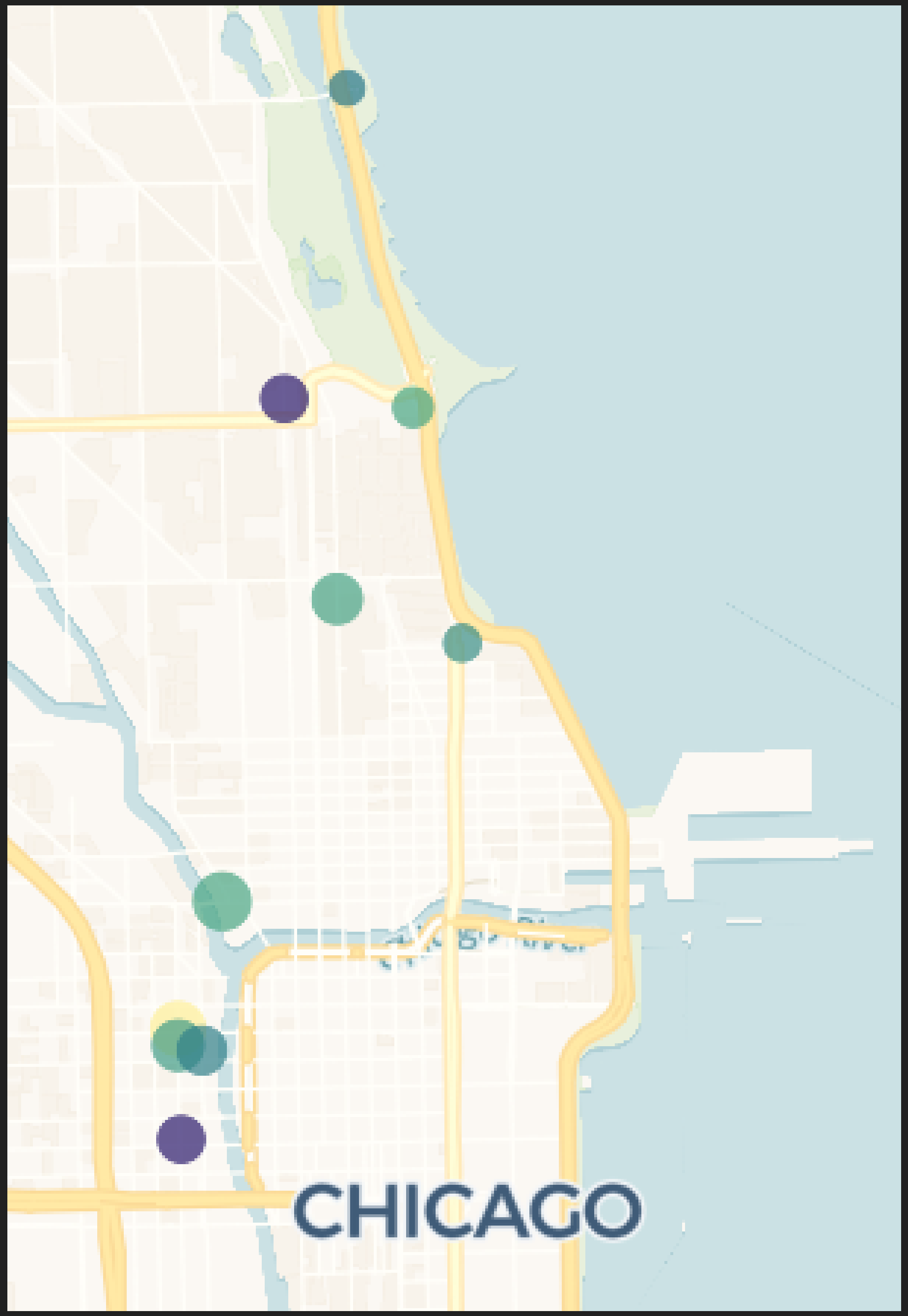
Casual riders rise gradually to a peak in the evening. Midday to evening is this group's active time.

TOP START STATION MEMBER VS. CASUAL

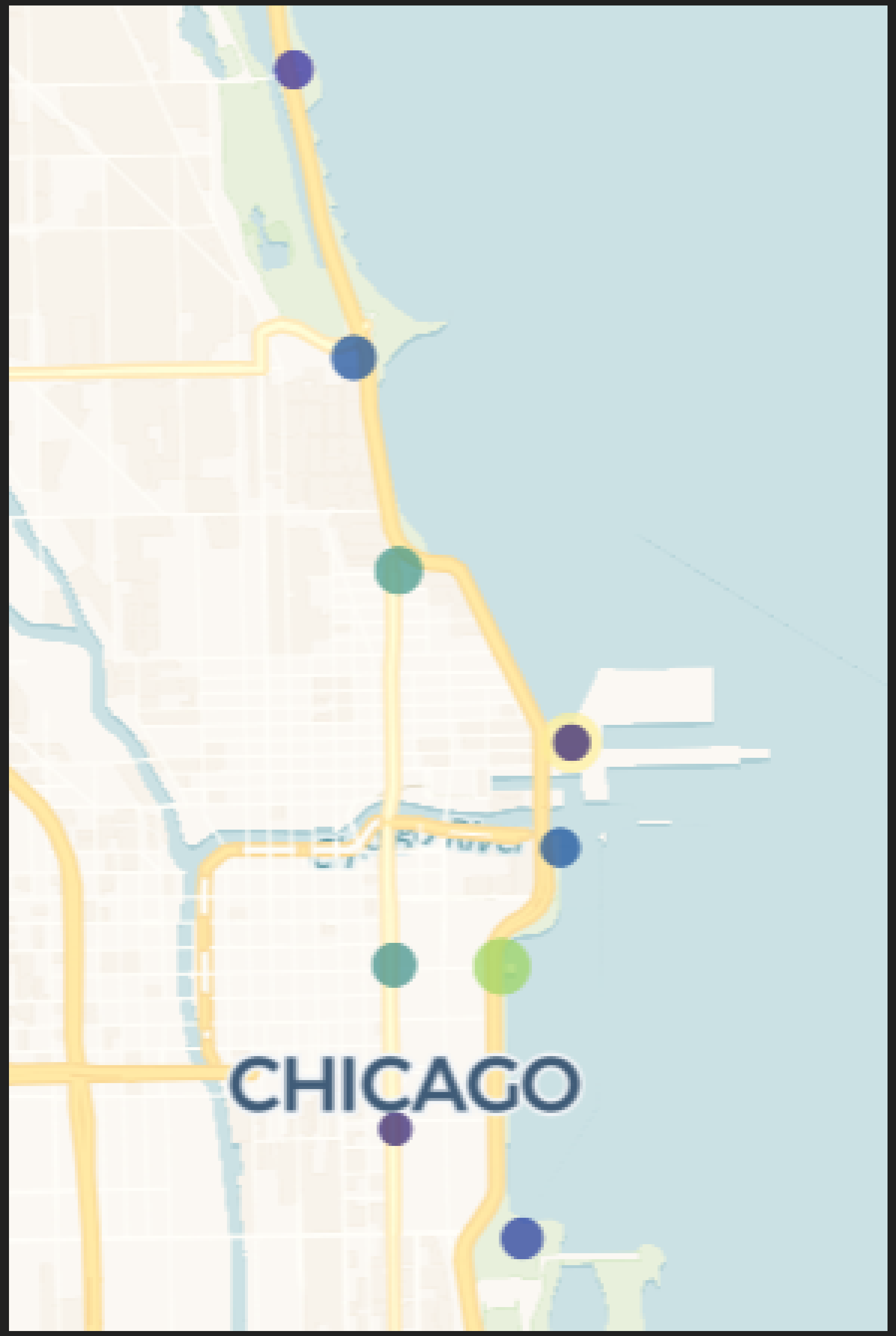
Map of the most popular start stations for both rider types.

Members usually start their trips in office and school areas in the city.

Casual riders start from tourist or leisure areas along the lakefront.

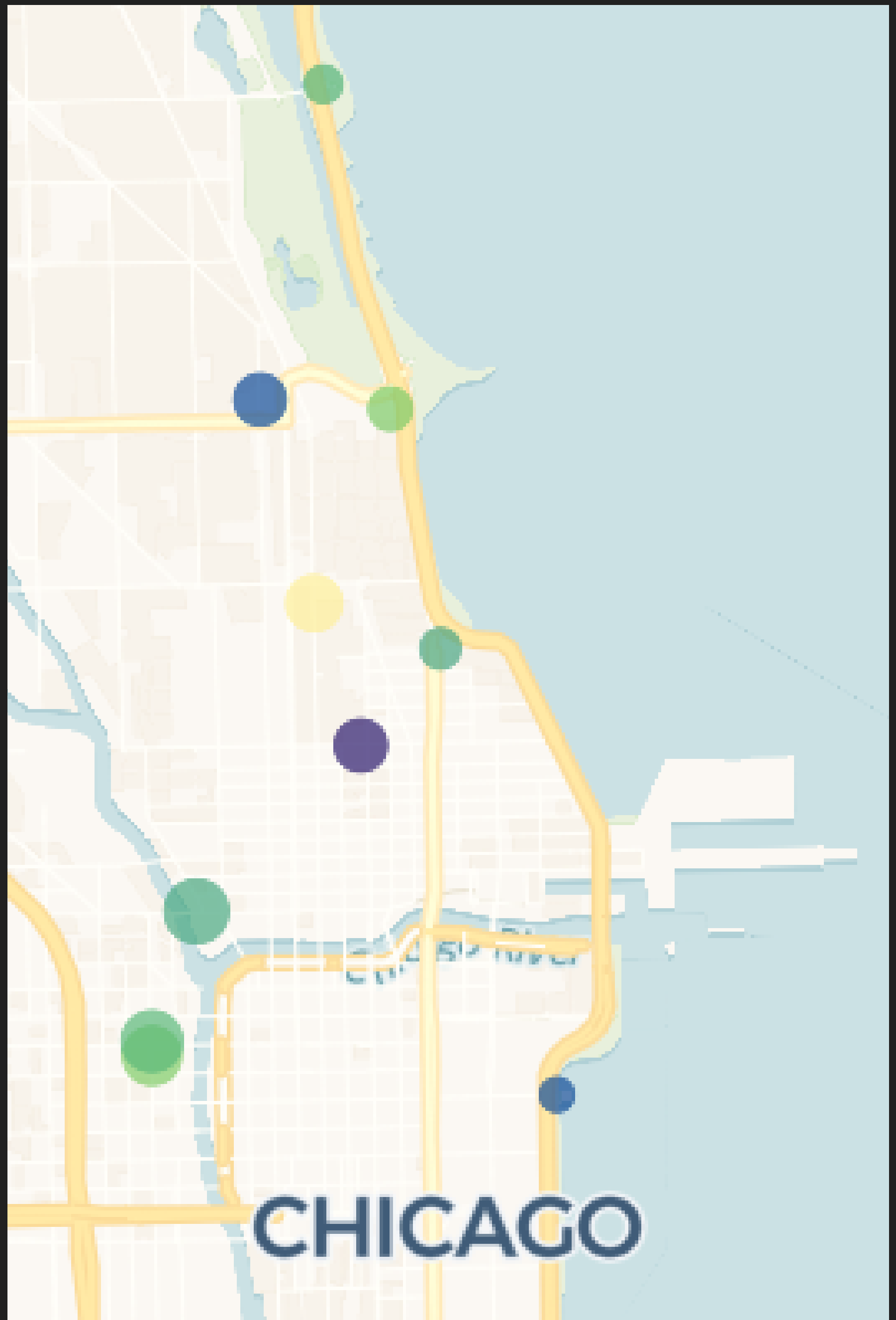


MEMBER



CASUAL

TOP END STATION MEMBER VS. CASUAL



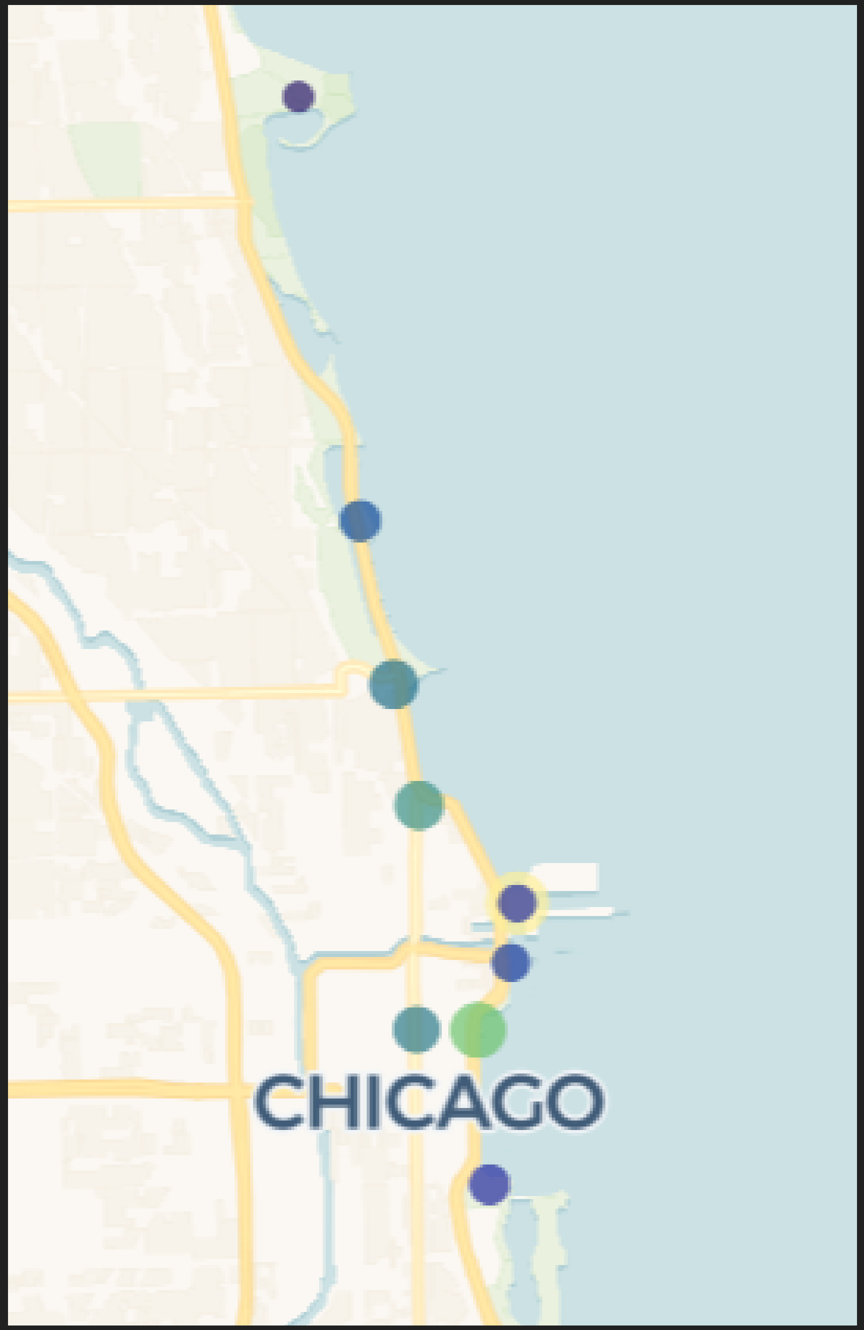
MEMBER

Map of the most popular end stations for both rider types.

This is similar to the start stations.

Members usually end their trips in office and school areas in the city.

Casual riders usually end in tourist or leisure areas along the lakefront.



CASUAL

KEY FINDINGS

How do annual members and casual riders use Cyclistic bikes differently?

Member	Casual
Rides more often in every season, but for shorter trips. They only ride longer when the weather is bad for outdoor activities.	Rides less often, but for much longer trips — mainly when the weather is good for being outside, from late spring to early fall.
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Most active at the start and end of the workday (rush hours). Rides more on weekdays (except Friday, which is slightly lower).	Most active from midday to evening. Rides more on weekends.

WHY WOULD A CASUAL RIDER DECIDE TO BUY A CYCLISTIC ANNUAL MEMBERSHIP?

- They ride often, or expect to ride often during a certain period, and have decided that a membership is better value than paying per ride many times.
- Convenience — cheaper access to bikes, or special member perks.
- A change in travel habits — after trying it, they start to see the chance to use bikes to avoid traffic during rush hour on workdays.

HOW CAN CYCLISTIC USE DIGITAL MEDIA TO ENCOURAGE CASUAL RIDERS TO BECOME MEMBERS?

Social media is strong for contextual and personalized marketing. We can use this, for example:

- Contextual & Location-Based Advertising : **Send ads to people in leisure or lakefront areas on weekends.**
- Retargeting based on Behavior : **If the system sees the same casual rider riding repeatedly within a month, send a Personalized Saving Report ad.**
- Seasonal Campaigns on Social Media : **A "Ride into Summer" campaign — a special discount for switching from casual to member in summer.**
- Content Marketing
- etc.

RECOMMENDATIONS

NEW AND DIVERSE
MEMBER OPTIONS

New, more flexible membership types — more varied to match riding habits, instead of just one annual plan.

- Shorter memberships, such as monthly or weekly.
- Season-only passes, such as Summer Pass, Ride from Spring, etc.
- Packages for specific times, such as a Rush Hours Package or a 12:00–18:00 Package.

CONTEXTUAL & SEASONAL
MARKETING CAMPAIGN

Build a contextual marketing strategy that uses rider behavior data combined with timing, season, tourist spots, holidays, or current events. This delivers offers that fit casual riders (Right Message, Right Time) and raises the conversion rate to membership.

COMMUTER INCENTIVE PROGRAM

Special perks for commuters, since this group is the main source of revenue.

For example, guarantee available bikes in office areas, and reward points for commuting that can be used for leisure rides, etc.

STRATEGIC BIKE & STATION OPTIMIZATION

Prioritize how bikes are distributed, since high-use stations clearly split between tourist spots and work areas.

For example, move bikes to the lakefront on Saturdays and Sundays, and back to business areas on Sunday evening.

FURTHER ADVANCED ANALYTICS

GHOST BIKE RECOVERY

Anomaly Detection

Find patterns in unusual trips (e.g. if a ride lasts over 24 hours and is left outside a station, could the bike be lost?).

Loss Prevention

Identify risk areas (hotspots) to build a real-time alert system. This lowers the cost of lost bikes and improves the recovery team's efficiency (Operational Efficiency).

FUTURE INVESTMENT

Spatial Optimization

Use geospatial data to find areas with high demand but not enough stations or bikes, to guide decisions on building more stations.

Targeted Expansion

Use data to decide where to add new docking points, add bikes on the right routes, and allocate budget for the highest ROI.

MARKET HEALTH AND EXTERNAL FACTORS

Predictive Modeling

Combine insights from external (3rd-party) data — such as market conditions, weather, fuel prices, or events — with ride data to support marketing, resource allocation, and future strategy.

Demand Forecasting

Build models to forecast demand in advance, so bikes can be distributed to the right places and prices can flex (Dynamic Pricing).